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UAV-TISP: A simulation platform for collaborative task training and algorithm development in multirotor UAV clusters

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ABSTRACT

To fulfill the training requirements for the daily operations of multirotor unmanned aerial vehicles (UAVs) clusters, a UAV cluster collaborative task integrated simulation platform (UAV-TISP) was developed. The platform integrates a suite of hardware and software to simulate a range of collaborative UAV cluster operation scenarios. It features modules for collaborative task planning, UAV cluster simulations, and tactical monitoring. The platform significantly reduces training costs by eliminating physical drone dependencies while offering a flexible environment for testing swarm algorithms. UAV-TISP supports both individual UAV and swarm operations, incorporating high-fidelity flight dynamics, real-time communication via user datagram protocol (UDP), and collision avoidance strategies. Utilizing the OSGEarth engine, it enables dynamic 3D environment visualization and scenario customization. Three key task scenarios—route flight, formation reconstruction, and formation transformation—were tested to validate the platform's efficacy. Results demonstrated robust formation maintenance, adaptive collision avoidance, and seamless task execution. Comparative analysis with Gazebo Sim revealed lower trajectory deviations in UAV-TISP, highlighting its superior accuracy in simulating real-world flight dynamics. Future work will focus on enhancing scalability for diverse UAV models, optimizing swarm networking under communication constraints, and expanding mission scenarios. UAV-TISP serves as a versatile tool for both operational training and advanced algorithm development in UAV cluster applications.

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1 Introduction

With the continuous advancement of research in the field of unmanned aerial vehicles (UAVs), drones have been widely adopted in various sectors. The integration of unmanned systems with manned systems enables collaborative tasks that cannot be achieved alone. This manned/unmanned collaboration is becoming essential in future defense systems, a topic that has been extensively discussed by aviation and defense researchers in recent years. Furthermore, the concept of drone-swarm cooperation, despite relatively new, can mitigate numerous risk factors, significantly reduce accidents, and lower costs if thoroughly validated in a virtual environment before implementation in real-world scenarios.

In Ref. [1], a manned-unmanned teaming (MUMT) training platform was proposed, where the ground control station (GCS) and payload operator (PO) can adapt to human-unmanned teaming scenarios in a simulation environment before engaging in actual operations. However, the main goal of this study was to enhance the PO's tactical skills via two-dimensional (2D) simulation rather than UAV cluster simulation. Reference [2] introduced an interactive simulation system for human-drone team coordination, allowing operators to communicate with drones through screen touch, voice, or multimodal inputs to improve task efficiency. Note that no actual flight dynamics were used for the simulation in this study or the three-dimensional (3D) environment simulation involved. Reference [3] presents a system for simulating tethered multirotor drones for indoor training in routine flight operations such as takeoff, hovering, landing, emergency procedures, and payload control. This system offers 3D visualization but excludes multidrone cooperative scenarios. Reference [4] mainly focused on the key technologies for constructing a simulation testing environment for drone swarms from the system perspective. It then primarily utilizes physical drones for training rather than relying on simulated models. Reference [5] introduced a drone simulation system that includes flight control simulation and 3D scene simulation, simplifying UAV flight dynamics calculations with linear models, and primarily focusing on single UAV operations.

Unlike previous studies, the UAV cluster collaborative task integrated simulation platform (UAV-TISP) can simulate both individual UAV and UAV swarms, allowing for the planning of tasks for a UAV swarm, and provides real-time visualization in a 3D environment, which can serve as an effective training tool for cluster collaboration and a simulation tool for the validation of cluster control algorithms. This research presents four main contributions: 1) The development of a simulation platform for collaborative tasks of rotorcraft UAV clusters, which enables inter-drone communication and integrates the simulation of task control, various UAV payloads, and routine and emergency operations of both single UAV and UAV swarms; 2) the creation of a diverse task scenario customization module that specifies multiple types of tasks for single drones or clusters to execute and monitor task data; 3) the use of the OSGEarth engine to simulate single UAV or UAV swarm flight states in a 3D environment; 4) the testing and demonstration of various multi-mode integrated tasks (such as route flight, formation flight, and formation reconstruction) using the proposed UAV cluster collaborative task simulation platform.

The remainder of this paper is organized as follows: Section 2 describes the functionalities of the collaborative task simulation platform modules. Section 3 discusses the key technologies and algorithms for drone control. Section 4 outlines the three scenarios used to test the collaborative task simulation platform. Finally, a discussion of the summary, conclusions, and future work of this research is provided.

2 Simulation platform design

2.1 Overall HW/SW implementation

This section introduces the overall structure of the UAV cluster collaborative task integrated simulation platform (UAV-TISP) and the functions of each component. UAV-TISP is divided into three main modules: UAV swarm simulation module, mission planning module (MPM), and visualization module (VM) [6]. The data flow between each module is shown in Fig. 1.

The UAV swarm simulator module (USSM) can simulate both individual drones and drone swarms. Each UAV is modeled as a multirotor copter and behaves according to a high-fidelity flight dynamics model. It was found that a typical tablet personal computer (PC) available on the market (e.g., NVIDIA P2200 and i7-10700

CPU) is capable of simulating up to 12 UAVs without compromising realistic animation. As mentioned before, each UAVs' state information (e.g., global position, altitude, velocity, pitch, yaw, and roll) is sent to MPM and VM via user datagram protocol (UDP) over Ethernet cables for high-quality visualization. They are also shared between the UAV swarm simulator and MPM for transmitting a control command.

A typical tablet PC serves as MPM as shown in Fig. 1. Via UDP messages over an Ethernet cable, MPM receives the UAVs state information indicated by the arrow coming into MPM from the UAV part. The UAVs' state (current position) information is then displayed and constantly updated on a 2D map with icons along with the VM state. Users can click these icons to check the state of each vehicle and select the UAV flight mode again. The flight mode selected by the users was then sent to the UAV swarm simulator. Owing to MPM, users can initiate, change, and monitor UAV activities during flight with almost no extra workload.

There is another display, the so-called VM, which is mainly responsible for providing a 3D visualization of the UAVs' state information, as indicated by the arrow coming into VM, while also being able to offer 2D scene monitoring, as shown in Fig. 1. An open-source OSGEarth platform was adopted for VM. VM shows the current positions of vehicles in a 3D simulation environment to the mission manager, who can easily monitor the cooperative operation of UAVs from various perspectives, which was missing in previous studies. The VM can become extendable and flexible in numerous senses, including visualizing high-quality 3D scenes or customizing simulation environments, such as adding obstacles. A tablet PC equipped with an NVIDIA RTX5000 and i7-1165G7 CPU was sufficient to support 3D visualization.

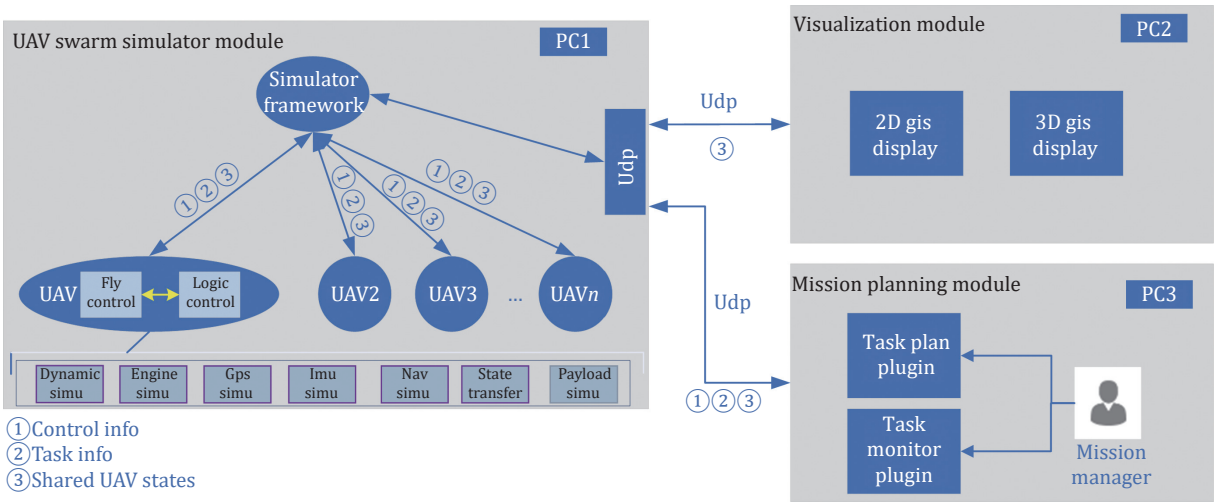


Fig. 1. UAV-TISP overall architecture.

2.2 *UAV swarm simulation*

The UAV swarm simulation module can respond to various commands for multirotor UAV control by simulating telemetry data for individual UAVs and UAV clusters. Each UAV object consists of two main modules: The fly control unit (FCU) and logic control unit (LCU). The LCU simulation module simulates payload-related data, whereas the FCU simulation module simulates flight telemetry data. The internal threads of the FCU and LCU modules run and implement data communication at the millisecond clock frequency. An integrated framework, fully developed by the authors in the C# programming language, receives guidance commands for UAVs (dynamical models) from the mission module using UDP messages over an Ethernet cable and also makes the UAVs communicate with each other.

To realistically simulate UAV flight, the FCU module was designed with the following functions: Dynamics, engine, inertial measurement unit (IMU), GPS, navigation, and state calculation. The dynamic simulation calculates the roll, pitch, and yaw accelerations in the world coordinate system by simulating engine rotation forces, ground reaction forces, and air resistance, thereby updating the UAV's position and velocity based on simulated time progression. Engine simulation can simulate manual and throttle modes, generate random speeds, and update the UAV state. The IMU simulation involves the numerical simulation of gyroscope values

in the X, Y, and Z directions, data validation, low-pass filtering, dual-axis compensation, and angular velocity calculation. The GPS simulation module simulates the position, velocity, and time (PVT) data of GPS and updates the E–N–U coordinates of UAV relative to a reference point in real time. The inertial navigation system (INS) simulation simulates the UAV's position and attitude data generated by the INS, and can correct them using data from the GPS simulation module. State transition calculations determine the UAV's current state, such as flight status, control mode, and fault status transitions. The LCU module primarily simulates UAV parachute landing, multivehicle formation control, and adjustment.

The state equation of the drone is as follows:

$$\dot{X} = s_i(x_i, c_i, u_i),\ i = 1, 2, 3, \cdots, N \tag{1}$$

where x_i represents the current state of the ith drone, c_i represents the adjustment amount of the drone calculated in real time by the flight simulation module, u_i represents the control information input by external users, such as flight path data and formation control data, and N represents the current number of drones in the simulation.

All UAV objects were created and managed using the integrated framework. The data processing workflow of the simulation framework is shown in Fig. 2. It is the integrated framework that receives the UAV number, initial positions, and flight routes of the UAV swarm from MPM and completes the initialization work. During the mission execution process, the framework receives UAV control commands from MPM and forwards information to each drone. When a swarm of drones is executing a mission, the formation control information and lead drone selection information are first sent to the framework, which then handles the information distribution. The state of each UAV is sent to MPM and VM using a UDP message at a frequency of 50 Hz.

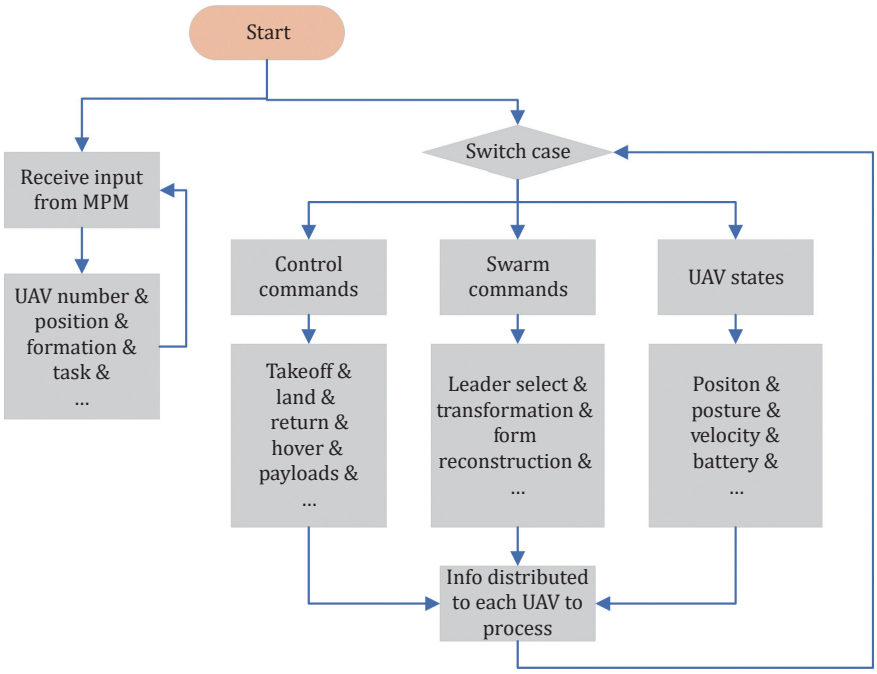


Fig. 2. Processing diagram of simulator framework.

2.3 Mission planning

MPM primarily performs tasks, such as drone cluster deployment, mission area planning, cluster formation planning, route planning, and action planning [7–10]. It can also simulate the control commands for individual units or clusters during drone flight. The architecture of the task planning module is illustrated

in Fig. 3. The task-planning module operates as a standalone executable program on a standard commercial computer and communicates with other modules via UDP or message queues.

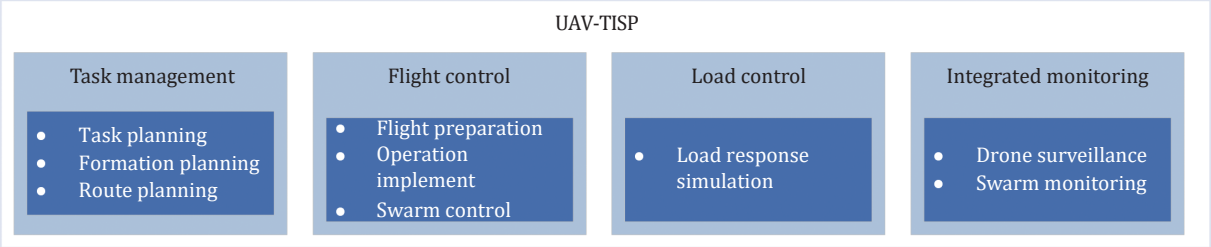


Fig. 3. Task planning module components.

Currently, mission planning categorizes tasks into two types: Attack missions and reconnaissance missions, each of which employs different planning models. The reconnaissance mission planning model calculated the reconnaissance area per UAV (W_o). The mission area (W) was then divided into W/W_o segments, followed by mission execution sequencing planning to produce the final reconnaissance mission plan. W_o is calculated using (2):

$$W_o = S_{\max} * w \tag{2}$$

where $S_{\max}$ represents the maximum range and w is the width of the reconnaissance field of view.

For attack mission planning, the required payload is determined by calculating the target dimensions, target value, and weapon lethal radius. Finally, the number of UAVs required to execute the mission was determined based on their carrying capacity.

Route planning primarily employs an improved A* algorithm, accounting for factors such as no-fly zones, waypoint fusion, and route smoothing.

2.4 Visualization

The mission planning module can plan tasks on a 2D electronic map, as shown in Fig. 4. In addition to the 2D map, the platform utilizes OSGEarth [11,12] to build 3D environments and simulate drone cluster flights

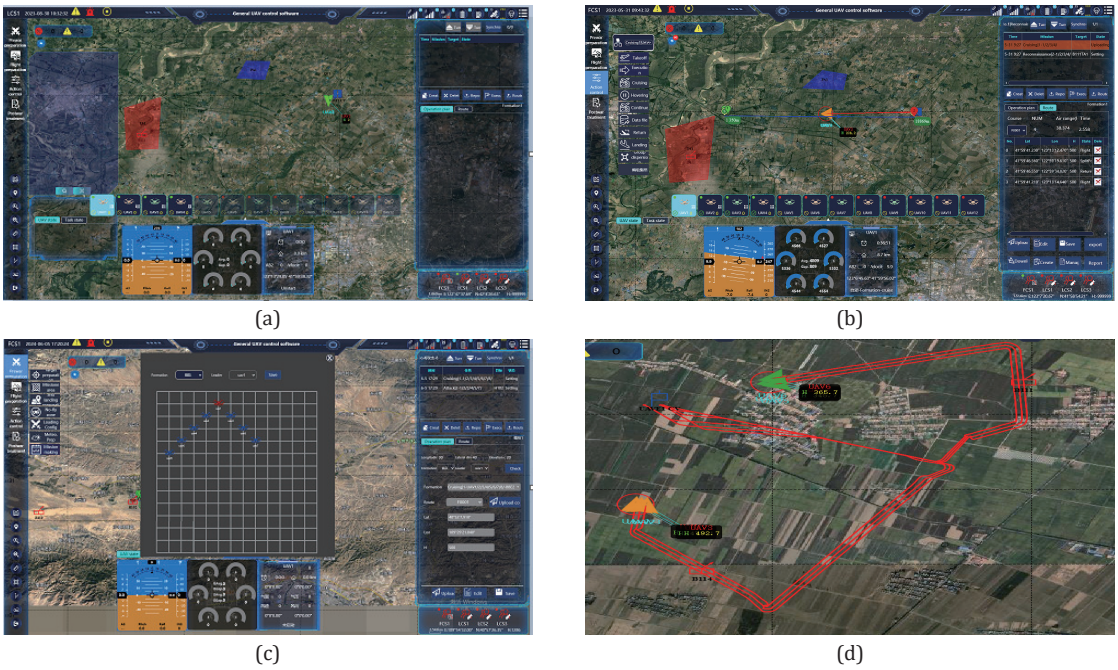


Fig. 4. Schematic diagram of mission planning software interface: (a) overall layout, (b) mission plannig, (c) formation planning and (d) flight path drawing.

in a 3D environment [13]. It is important to note that the 3D simulation platform serves solely as a display method and can be replaced with other platforms such as Unity3D or Unreal, as needed. OSGEarth is a geographic information system (GIS) software primarily used to create and present geographic data and spatial information on the Earth's surface. It includes GISs, environmental simulation, and weather simulation, allowing users to interactively browse the Earth's surface, add geographic markers, and perform terrain analysis. The 3D scene simulation is shown in Fig. 5.

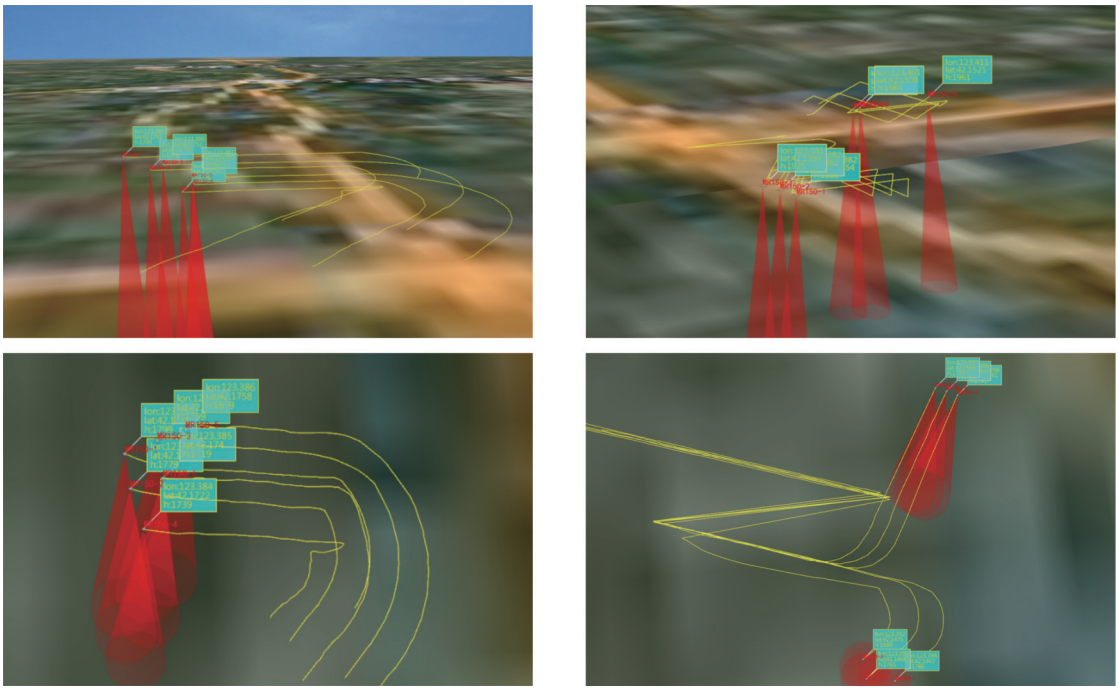


Fig. 5. 3D visualization.

3 Key strategy design and implementation

This section introduces three core strategies involved in the execution of drone swarm missions: Formation maintenance, formation collision avoidance measurement, and formation turning. The entire process encompasses the intelligent control of cluster formation, maintenance, restructuring, and collision avoidance, enabling the coordinated execution of multiple tasks by the swarm.

3.1 UAV formation maintenance strategy

A spatial Cartesian coordinate system (0-*XYZ*) was established based on the cluster formation. Assuming that the flight route is a continuous straight line in a 3D space parallel to the *X*-axis, the position of each point on the route can be precisely calculated. The formation maintenance strategy involves the leader aircraft identifying the flight route and adhering to it as closely as possible, whereas the follower aircraft focuses solely on following the leader without actively seeking the route [14].

Using the leader aircraft as the center, a virtual search sphere with radius d was created. The position of this sphere relative to the flight route was analyzed by solving the simultaneous equations of the search sphere and route.

$$\begin{cases} (x-a)^2 + (y-b)^2 + (z-c)^2 = d^2 \\ y = l \\ z = h \end{cases} \tag{3}$$

where a, b, and c are the leader aircraft coordinates, and l and h are the Y-axis coordinate and height of the route, respectively, solving for the unknown variable X.

There are two possible situations, as shown in Fig. 6, where an intersection exists and one where it does not. When there is an intersection, the point with the larger X-axis coordinate is chosen as waypoint Q towards which the leader aircraft adjusts its direction to fly. If there is no intersection, the projection point of that unit on the route is selected, and the X-coordinate of that point plus d becomes waypoint Q.

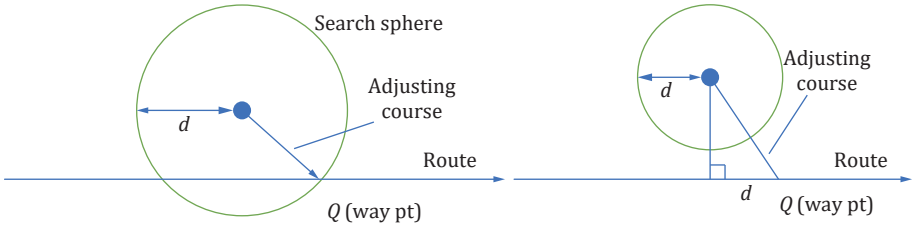


Fig. 6. Schematic diagram for lead aircraft route search.

3.2 Drone formation conflict avoidance strategy

There are two forms of formation conflict avoidance: One in which the entire formation avoids conflicts, with the lead drone executing the avoidance strategy while the wingman follows the lead's movements, and the other when there is collision risk within the formation due to disturbances, where the lead drone maintains its course while the wingman performs avoidance maneuvers based on the strategy [15].

Beginning with oneself as center D, a search sphere with radius kd is defined, where k is a suitable coefficient. When a target enters the search sphere ($s < kd$), avoidance mode is activated. Taking target O as the center, a ball with radius $(k-1)d$, called the adjustment ball, is defined. The intersection of the ball with plane T at the cruising altitude is a circle C, with the distance from O to T being l. The radius of circle C is $\sqrt{((k-1)d)^2 - l^2}$, with its center being the projection of O onto T. As T is a plane parallel to the XY plane, the coordinates of the center of circle C are the X and Y coordinates of the adjustment ball O. Circle C represents the avoidance circle. The relationship between search sphere and avoidance circle is shown in Fig. 7.

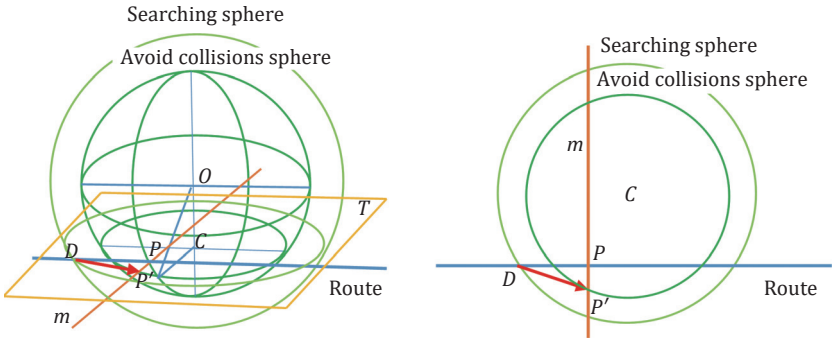


Fig. 7. Schematic diagram of formation collision avoidance strategy.

Considering the relation between the beacon point P calculated without other targets within the plane of route T and the vertical line m through the avoidance circle C, if there is no intersection point, it is known that the distance between oneself and the target is large enough (at least greater than $(k-1)d$) to not require adjustment and maintain the previous beacon point. If there is an intersection point, the beacon point P' is chosen with a higher Y coordinate, always maneuvering around the target's right side. If the two formations are at risk of collision, adopting the same strategy will result in a formation resembling a roundabout at the time of encounter, allowing the two formations to pass through the roundabout in succession before returning to their own course.

3.3 UAV formation turning control measurement

The turning process was designed to be slow in Ref. [16]. Fig. 8 shows two possible turning points, judging whether to switch the route by determining a circle with a radius of d as the current route endpoint

and whether the aircraft has flown past the route waypoint N in its own route coordinate system. When an aircraft enters the rejection zone, the route is switched, changing the target route to the next route segment. There are two turning modes: Turning through waypoints to ensure waypoint passage, and early turning to ensure smooth route transitions. Waypoints N_1, N, and N_2 formed a turning route, with the orange area as the rejection zone, the green line as the early turning route, and the red line as the turning route after passing waypoints.

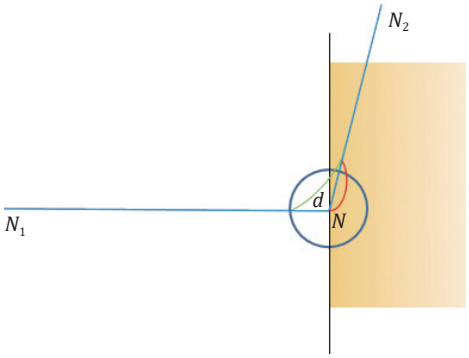
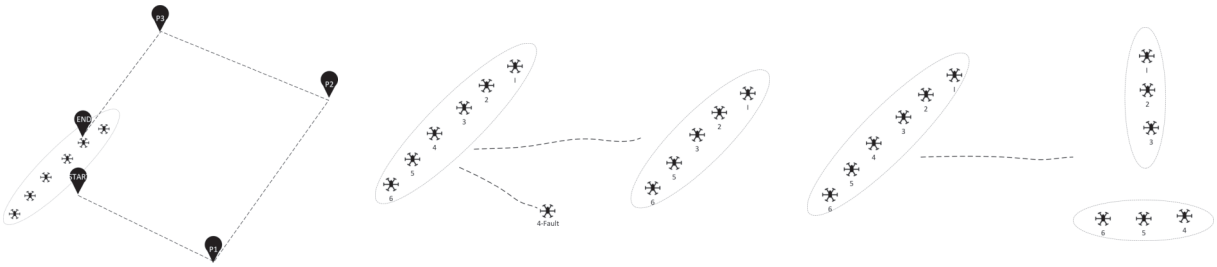


Fig. 8. Schematic diagram of turning route switching.

4 Validation analysis

Drone swarms face numerous challenges during real-world training, including environmental complexity, cost, safety risks, and coordination issues. In contrast, simulation environments provide a safe, efficient, and economical method to conduct validation and training. In a simulation environment, the potential risks associated with real operations, such as drone collisions, equipment damage, or personal injuries, can be avoided. This allows researchers to boldly experiment with various strategies and algorithms, without facing actual safety threats. Testing in the real world requires significant investments in human resources, material assets, and time. However, simulations can reduce these costs, especially for large-scale drone-swarm operations. Simulations allow for quick testing and design iterations without the need to wait for hardware modifications or field tests, thereby accelerating the development of drone algorithms and systems. Simulation runs can generate large amounts of data, facilitating in-depth analysis of the behavior and performance of drone swarms, which can optimize algorithms and strategies. Different operational scenarios, task complexities, and swarm behaviors can be easily created and tested in a simulation environment, thereby enhancing the adaptability and robustness of the models.

In this section, three task scenarios were designed to evaluate the proposed UAV-TISP: Route flight, formation reconstruction, and formation transformation, as shown in Fig. 9. The flight parameters for the UAVs are listed in Table 1. These three scenarios were selected primarily because they encompass the key characteristics and operational capabilities of drone swarms. Route flight is the most fundamental operation of a swarm, involving the basic navigation and stable flight capabilities of drones. It serves as the core for many drone missions such as patrolling, monitoring, and data collection. The formation change demonstrates the ability of drones to adjust their flight arrangements when performing specific tasks, making it a direct representation of swarm collaboration. Formation reconstruction shows the swarm's dynamic adaptability, allowing for flexible adjustments in response to different environments or task requirements. From the aspect of multitask adaptability, route flight is suitable for search and rescue, formation change is applicable for real-time monitoring, and formation reconstruction aids in avoiding obstacles or hostile threats. Finally, all three scenarios emphasize the collaborative capabilities of drones. Each drone must share information in real time to achieve coordinated flight and maintain proper formation. These scenarios form the basis for



(a) (b) (c)

Fig. 9. Task scenarios schematic diagram: (a) route flight demonstration, (b) formation reconstruction demonstration, and (c) formation transformation demonstration.

studying centralized and distributed control algorithms. Finally, a comparison between UAV-TISP and Gazebo Sim was conducted using the route flight scenario to verify the capability and performance of UAV-TISP.

Table 1
Route and formation parameters.

Parameter	Value	Definition
Formation longitudinal length	30	During cluster drone flight, a slanted vertical formation needs to be maintained.
Formation lateral length	40	
Collision avoidance radius	30	UAV collision avoidance rules parameter
Swarm leader	UAV1	The leading drone of the formation is defined.

The formation of six drones was used to establish a flight route and control the takeoff and flight of the drone cluster through the simulation platform. During flight, the simulation platform can simulate drone failures to verify the formation maintenance capabilities of the drones. Simultaneously, formation change commands can be issued to control the drones to split the group formation into two separate formations. Each drone was set with attributes UAV1, UAV2, UAV3, UAV4, UAV5, and UAV6, with UAV1 designated as the leader [17]. The initial formation was a column with a vertical spacing of 30 m, a horizontal spacing of 40 m, and a collision avoidance distance of 30 m.

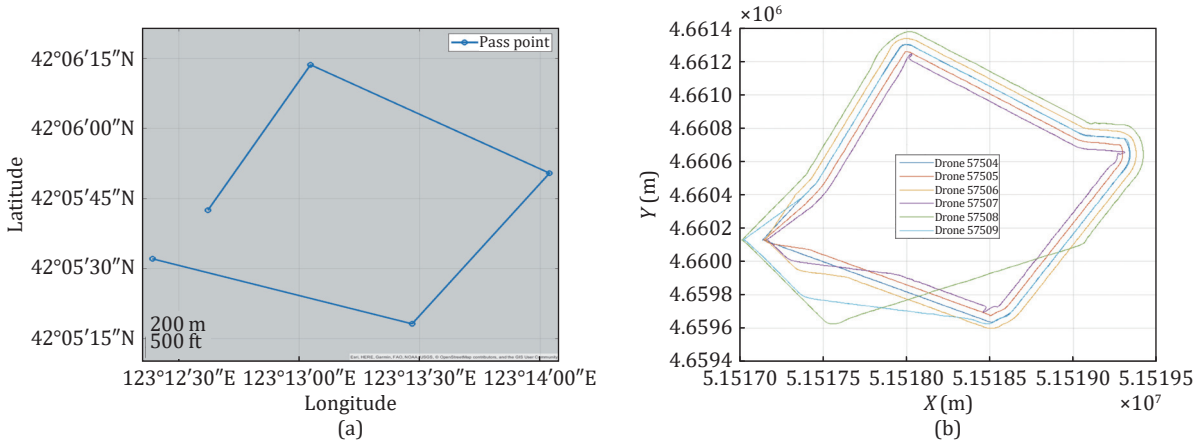
4.1 Route flight

During the route flight process, a cluster of six UAVs flies in a specific formation to complete the entire sequence of takeoff, route flight, and landing according to the route and formation data. In the formation takeoff phase, all drones take off simultaneously and hover at 30 m awaiting the next command. Once the flight command is received, the flight is executed according to specified vertical, horizontal, and altitude differences. The lead drone follows the route, whereas the follower drones maintain their formation based on the given parameters. Once formation is established, the entire flight process adheres to the required formation.

Fig. 10(a) illustrates the route of the drone formation flight, including five waypoints specified by latitude and longitude. Figs. 10(b) and (c) show the 2D and 3D simulations of the formation flight process, from hovering to final landing. Fig. 10(d) shows the altitude changes during the formation flight, highlighting the maintenance of a distance of approximately 30 m between adjacent drones to avoid collisions.

4.2 Formation reconstruction

To verify the ability of the UAV swarm formation reconstruction, six UAVs (UAV1, UAV2, UAV3, UAV4, UAV5, and UAV6) were used to fly in formation and simulate a scenario in which UAV3 experiences a fault during flight, as shown in Fig. 11. The remaining UAVs adjust their formations and continue flying according to the original formation, maintaining a minimum separation distance of 30 m during flight. Figs. 11(a) and



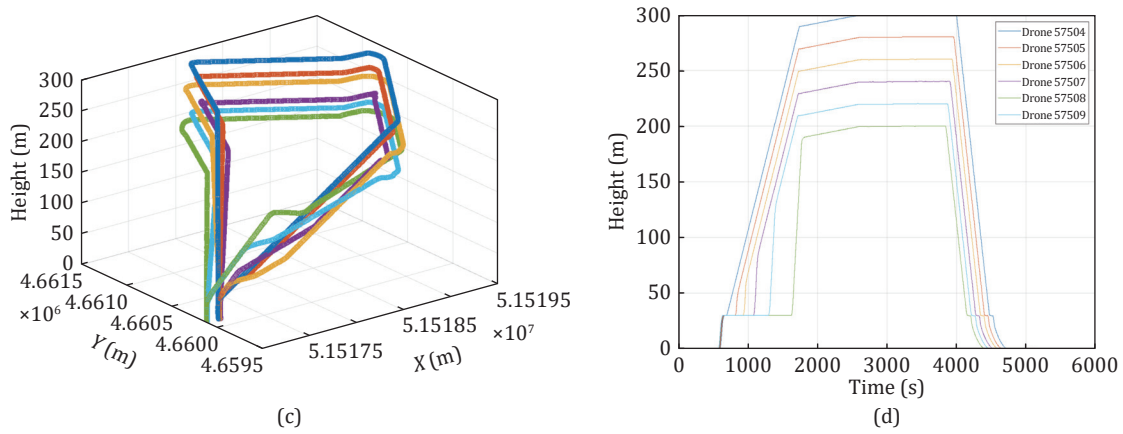


Fig. 10. Quantitative evaluation of performed mission route flight: (a) cluster fight route, (b) 2D simulation results of formation flight, (c) 3D simulation results of formation flight, and (d) altitude variation during flight.

(b) show the 2D and 3D simulation results of the formation reconstruction process. From Fig. 11, it is evident that UAV3 malfunctions with abnormal telemetry data, while the other UAVs complete the entire process of formation reconstruction. Fig. 11(c) illustrates the variation in flight altitude for each UAV during the formation flight process, demonstrating that the height between adjacent UAVs within the formation is maintained at 30 m. By analyzing the feedback telemetry data from the UAVs after reconstruction, the

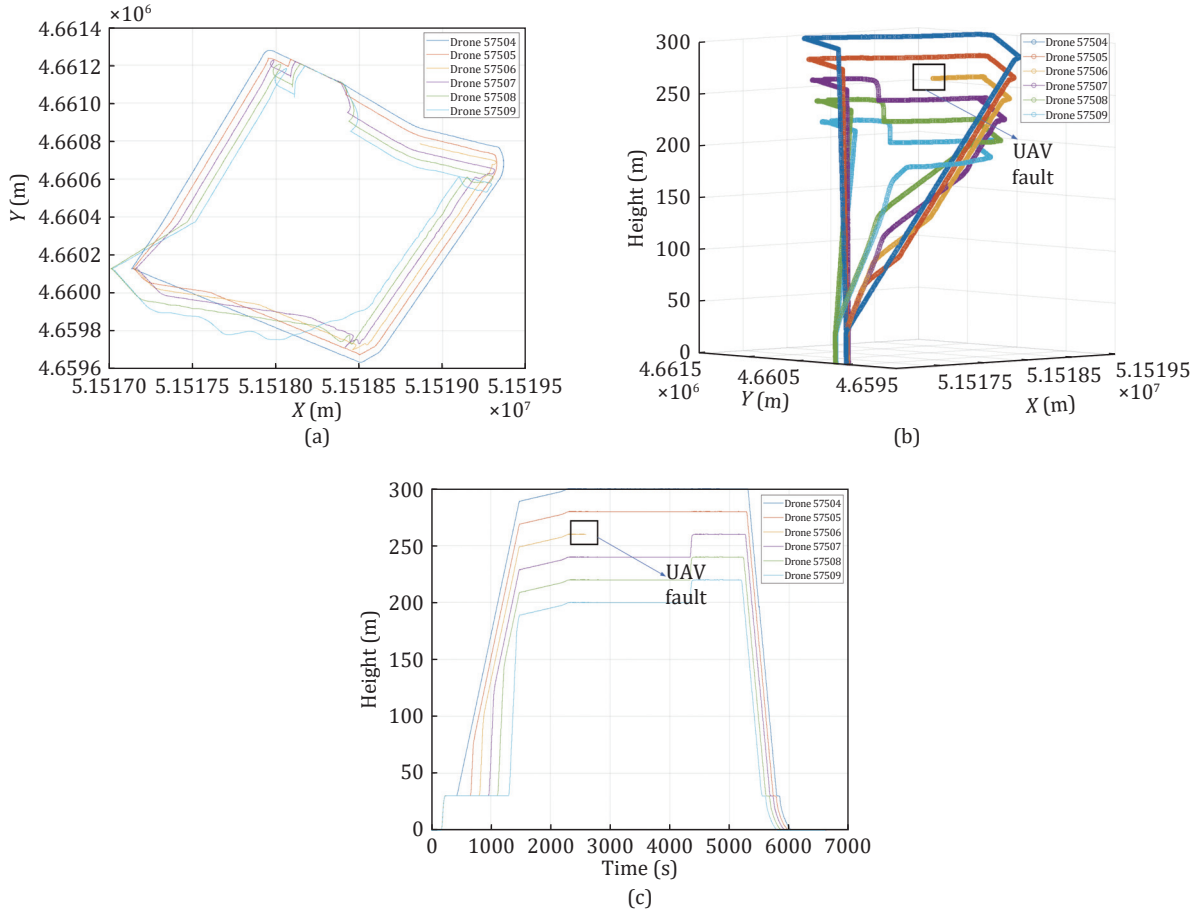


Fig. 11. Process of formation reconfiguration for UAV swarm: (a) 2D simulation results of formation reconstruction, (b) 3D simulation results of formation reconstruction, and (c) altitude variation during flight.

position information of the UAVs can be visualized, thereby fully verifying the feasibility and reliability of the formation maintenance strategy in this design.

4.3 Formation transformation

To verify the ability of UAV swarm formation splitting and transformation, a fleet of six UAVs (UAV1, UAV2, UAV3, UAV4, UAV5, and UAV6) was organized into a formation. Initially, the fleet cruises along a route during the wedge formation. A command is then given to disband the formation and upload new route data to the UAV swarm, thereby splitting the original group into two new formations, as shown in Fig. 12. Formation one consists of UAV1, UAV2, and UAV3, with UAV1 as the leader, flying in a line formation with 30 m vertical spacing, 40 m horizontal spacing, and 20 m altitude difference. Formation two consists of UAV4, UAV5, and UAV6, with UAV4 as the leader, flying in a staggered formation with the same parameters as formation one.

Figs. 12(a) and (b) depict the 2D and 3D simulation results of the formation transformation process, showing the UAV swarm hovering at 30 m, flying along the route, and splitting into formations. Fig. 12(c) illustrates the altitude variations of each aircraft during the formation flight process, revealing that the adjacent UAVs maintain a 30 m vertical spacing within a formation. Prior to formation reconstruction, the six UAVs flew in a wedge formation along the route. The formation was restructured into two new formations by executing the disband command during the flight. The UAV simulation module can execute formation splitting according to the new mission data, and the reconstructed positions of the UAVs can be observed through the feedback of telemetry data, effectively validating the feasibility and reliability of the formation maintenance strategy in this design.

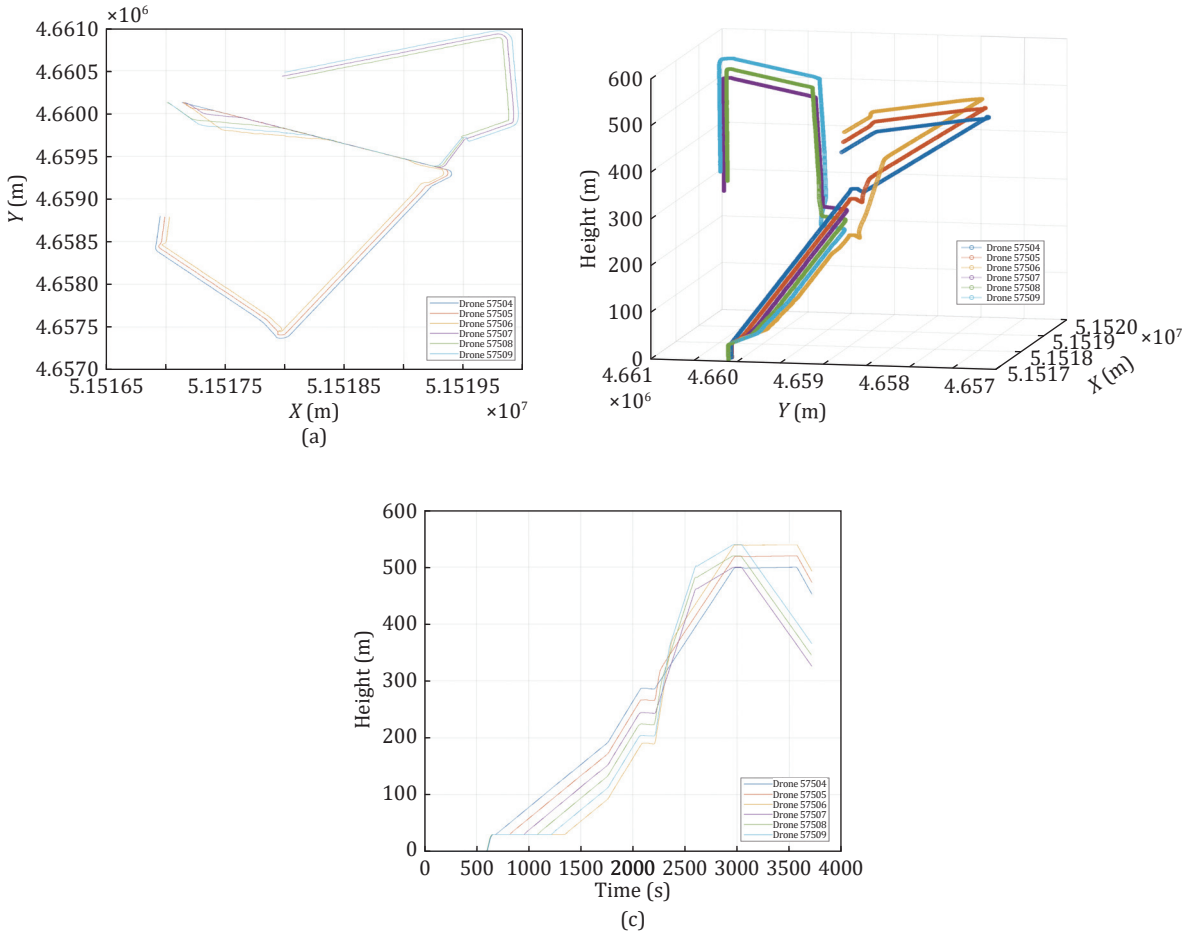


Fig. 12. Process of formation transformation for UAV swarm: (a) 2D simulation results of formation transformation, (b) 3D simulation results of formation transformation, and (c) altitude variation during flight.

4.4 Comparative analysis

To verify the capability of UAV simulation of UAV-TISP, a comparison was conducted between UAV-TISP, Gazebo sim, and actual flight data using a route flight scenario. Gazebo is a powerful robotic simulation environment that allows for the development, testing, and visualization of robotic systems in a 3D virtual space. In the route flight scenario, one UAV flew along a straight line.

Figs. 13(a) and (b) depict the 2D and 3D simulation results, showing one UAV flying along the route. The 2D and 3D simulation results show that the flight trajectories of UAV-TISP, Gazebo sim, and the actual flight data are very similar. The flight paths of the three platforms nearly overlapped, implying that the proposed UAV-TISP can offer the expected simulations.

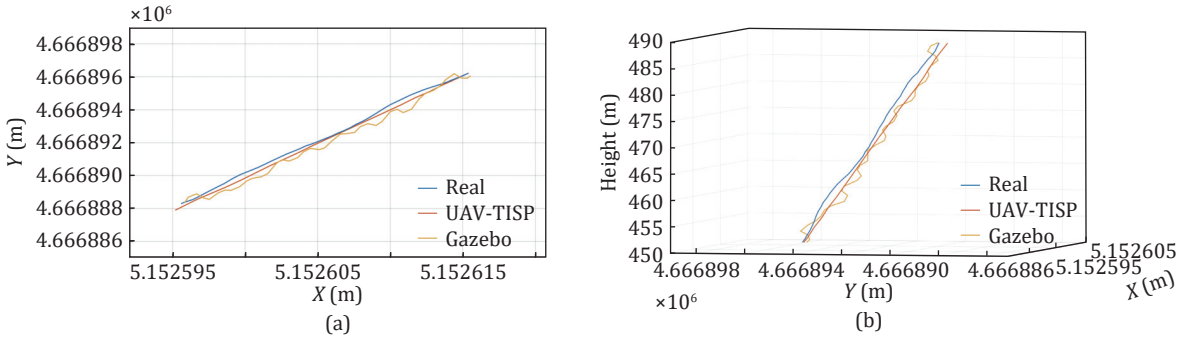


Fig. 13. Route flight comparison of UAV-TISP, Gazebo, and actual data: (a) comparison of 2D data and (b) comparison of 3D data.

To further analyze the differences among the three types of data, the errors of the UAV-TISP coordinates and Gazebo coordinates relative to the actual flight data were calculated separately. The Euclidean error, defined as $||\Delta x+\Delta y+\Delta z||$ of the coordinate errors, was also calculated for comparison.

The data in Fig. 14(a) show the deviation of the simulation platform and Gazebo compared with the actual flight data. It can be observed that as time progressed, the errors of both algorithms relative to the actual records continued to increase. However, the error for Gazebo was larger, whereas the simulation platform discussed in this paper exhibited a more gradual increase. Both platforms showed considerable errors in the *X*-direction, whereas errors in other directions were relatively small. As the general trend of the flight route is along the *X*-axis, this indicates that the simulation platform has a greater error in the route direction and a smaller error in the non-route direction. In addition, it is evident that the deviation of the simulation platform from the actual data is smoother, whereas Gazebo's deviation is more variable.

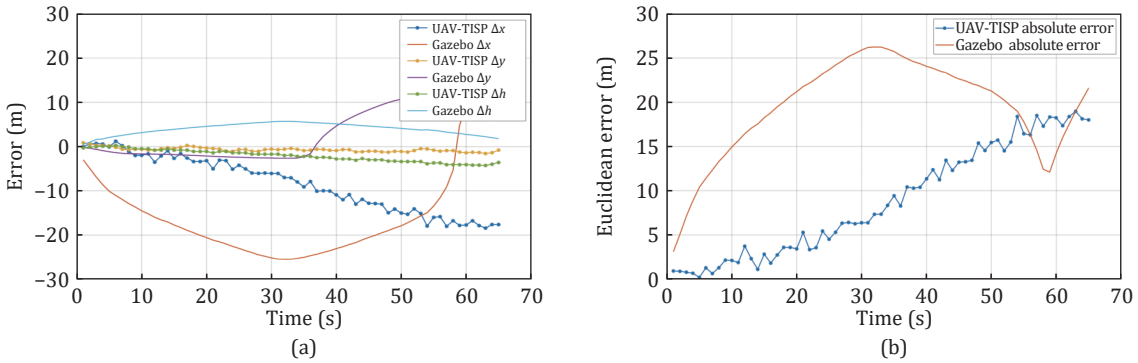


Fig. 14. Error comparison of UAV-TISP, Gazebo relative to actual data: (a) coordinates error comparison and (b) euclidean error comparison.

The data in Fig. 14(b) represent the absolute errors in the coordinates. It can be concluded that the accuracies of the two platforms are similar, and that the accuracy of UAV-TISP is superior to that of Gazebo in the partial flight process. However, the algorithm error becomes more pronounced as time increases, indicating that there is room for improvement in this area, such as improvements in handling extreme values.

5 Concluding remarks and future work

In this study, we proposed a platform called UAV-TISP, which includes drone cluster simulation software, cluster mission planning software, and 2D/3D environment display software. The UAV-TISP platform can implement algorithms for drone formation maintenance, collision avoidance, and restructuring. Operators interact with the UAV cluster through mission planning software to execute tasks and control commands, closely monitoring cluster flight situations through 2D and 3D monitoring interfaces. The proposed UAV-TISP is tested under multiple mission scenarios. The simulation platform can serve as a training tool for drone clusters, significantly reducing the training and support costs by eliminating the need for physical drones and data linkages. Future work can focus on studying the universality and scalability of drone cluster simulations and mission planning software to accommodate more drone models and mission scenarios. In addition, in the case of poor network communication when the swarm is subject to interference, it is meaningful to attempt to optimize the routing and networking mechanisms of the wireless mesh network (WMN) [18] formed by the swarm to enable each node within the swarm to maintain communication.

CRediT authorship contribution statement

Zhi-Bin Xie: Methodology, Design, Investigation, Research, Writing–Original draft. **Hong-Ying Zhang:** Supervision, Validation, Visualization, Project administration. **Xiu-Qin Cheng:** Methodology, Formal analysis, Investigation, Visualization. **Ze-Kun Wang:** Investigation, Simulation, Test, Writing–review & editing, Theoretical analysis. **Xiao-Long Wang:** Methodology, Simulation design of UAV swarm, Data processing, Writing–review & editing. **Kai Xi:** Methodology, Visualization, Formation strategy design, Algorithm verification, Writing. **Qiang Guo:** Investigation, Test, Mission planning module design, Writing–review & editing.

Declaration of competing interest

The authors declare no conflicts of interest.

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